PHASE REPORT • MHD-WAAM THESIS • PHASE 2

Materials and Scaling

The property set, the number the source paper never needed, and a scaling result that moves the thesis

Sabbir Dipto | Al–5%Mg (5A05 substrate, ER5356 wire) | replication target: Zhao, Wei, Long, Chen, Liu & Ou (2021), Welding in the World 65:1571–1590

THE ONE-PARAGRAPH VERSION

Phase 2 was meant to be bookkeeping — put the paper's Table 4 into a dictionary, add electrical conductivity, check the units. It did that. But working the force scaling by hand turned up something that changes the plan: **the magnetic module in laserbeamFoam is in the wrong physical regime for this problem, by two to three orders of magnitude, at every field strength anyone applies.** Not slightly off — structurally unable to show the effect. That upgrades welding-current injection from the thesis's novel contribution to the step everything else waits on. It also explains, cleanly and in advance, why Phase 1 saw no magnetic effect on the melt.

1. What Phase 2 set out to do

The roadmap's Phase 2 has four deliverables: get Zhao's Table 4 into an OpenFOAM dictionary with real provenance rather than copied numbers; chase the temperature-dependent properties back to their source rather than guessing; add the electrical conductivity the paper never needed; and work the force scaling by hand so the crossover field is yours and not inherited from an earlier draft of the plan.

All four are done. The output is five files:

FILE	WHAT IT IS
alMg5Properties	OpenFOAM dictionary. Every entry carries a provenance tag.
alMg5_properties.csv	Property tables, 300–2000 K in 25 K steps.
make_props.py	Generates both of the above from the fits. Nothing is typed by hand.
scaling.py	The force-scaling analysis. Change an input, watch what moves.
scaling_report.txt	Its output, captured.

WHY EVERY VALUE CARRIES A TAG

The dictionary uses four provenance tags, and they are the point of the file. **[ZHAO]** printed in the paper. **[MEAS]** an independent measurement, source named. **[CONSTR]** a construction from measured data, method named. **[EST]** an estimate whose sensitivity must be tested.

A number you cannot defend in a viva is worse than no number. Tagging forces the distinction at the moment of writing, when it is cheap, rather than at the moment of questioning.

2. The headline: two forces share one name

"Lorentz force" names two couplings that behave completely differently, and the whole thesis turns on which one is operating.

	STIRRING	BRAKING
Where the current comes from	The arc — 135 A, injected	The melt's own motion, sigma(u × B)
Scaling with field	Linear in B	Quadratic in B
What it does	**Drives** flow	**Damps** flow
Needs a current source?	Yes	No

At the melt pool the arc's current density is $|J| = I / (\pi \cdot \sigma_j^2) = 1.07 \times 10^7$ A/m^2. Comparing the two couplings as stresses over the pool scale:

FIELD (mT)	STIRRING (Pa)	BRAKING (Pa)	RATIO
10	322	0.33	**981 ×**
20	645	1.31	**491 ×**
50	1611	8.21	**196 ×**
100	3223	32.85	**98 ×**

*The two cross only at **9.8 tesla** — an order of magnitude beyond any laboratory magnet, let alone any welding experiment.*

WHAT THIS MEANS FOR THE SOLVER

laserbeamFoam's MTHD module contains only the braking term. It is a laser code: there is no welding current, so J = curl(H) is purely motional. At 20–50 mT — the range the experimental literature actually uses — it is therefore modelling a force that is 200 to 500 times smaller than the one that dominates.

This is not a refinement to add later. Without current injection the model cannot show the effect *at all*: not weakly, not at higher field, not on a finer mesh. It is in the wrong regime. **Phase 6 stops being the novel result and becomes the load-bearing step.**

3. Why one current brakes and the other stirs

These are not two different forces. Both are J × B — the same formula and the same physics. The entire difference is **whether J depends on the velocity u**.

The injected current does not know the melt exists

The power supply drives 135 A through the pool. That current is fixed by the external circuit, not by what the liquid metal is doing. So the force

```
F = J_arc x B
```

is a **fixed vector** once B is chosen. It points somewhere definite and pushes. It is an applied force, in the same mathematical family as gravity: it sits on the right-hand side of the momentum equation as a source, and it **does positive work on the fluid**. The energy comes out of the welding power supply.

The motional current is made of u, so it can only fight u

With nothing injected, current exists only because the melt moves through the field. Charges in a moving conductor feel q(u × B), which drives a current

```
J_ind = sigma (u x B)
```

Substituting that back into J × B and expanding the triple product with (a × b) × c = b(a · c) - a(b · c):

```
F = sigma (u x B) x B = sigma [ B (u . B) - u B^2 ]
```

Nothing so far assumes anything about direction — that expression holds for any B and any u. To read it, split the velocity into a part along the field and a part across it, u = u_par + u_perp. Since u_perp is perpendicular to B, the dot product sees only the parallel part, and the whole thing collapses:

```
u . B = |u_par| B

F = sigma [ B^2 u_par - B^2 (u_par + u_perp) ] = - sigma B^2 u_perp
```

Antiparallel to the velocity, with a minus sign that never goes away. This is not a force with a direction; it is a drag coefficient. It is algebraically identical to an extra viscosity, which is why it belongs on the *left*-hand side of the momentum equation as a sink rather than the right as a source.

The field direction does not change the law, only what it acts on

The result above is completely general, and the u_perp in it is the whole story of field orientation. Flow *along* the field feels nothing at all — not weak damping, none — because J = sigma(u × B) cannot see the parallel component: u_par × B is identically zero. Physically, you cannot generate a motional EMF by sliding along a field line. Charges are only deflected when they cut across field lines.

Written as an operator, that makes braking a projection:

```
F = - sigma B^2 ( I - Bhat Bhat ) . u
```

> **BRAKING IS ANISOTROPIC, AND THAT IS NOT A DETAIL**
>
> The bracket projects onto the plane perpendicular to B. So magnetic braking is **not** a scalar drag like viscosity — it is a drag with a hole in it, and the hole points along the field.
>
> This is why strong-field MHD flows become quasi-two-dimensional: the field damps everything except motion along itself, leaving columnar structure aligned with B. The flow is not stopped so much as *flattened into the field direction*.

For this melt pool that means field orientation remains a real experimental variable even for pure braking — it selects which of Zhao's circulations survives:

FIELD	DAMPS	SURVIVES
B along y, transverse	u_x and u_z — **the main Marangoni recirculation loop**	Sideways spreading
B along z, vertical	u_x and u_y — horizontal spreading	The vertical plunge under the falling droplet
B along x, travel direction	u_y and u_z — the transverse circulation	Longitudinal flow

Which sharpens the contrast with stirring: field direction sets the direction of a new force for stirring, but sets which components survive for braking. Both depend on orientation; they depend on it differently.

WHY THE SIGN CANNOT COME OUT ANY OTHER WAY

Making a current costs energy — you dissipate J^2/sigma as Joule heat. In the injected case the power supply pays that bill. In the motional case there is no supply, so it is paid out of the flow's own kinetic energy. The melt must slow down.

This is Lenz's law. If motional current ever accelerated the fluid, you would have a flow that speeds itself up while simultaneously heating itself, out of nothing.

The analogy is a motor versus a brake. Supply current to a conductor in a field and you have a motor. Let a conductor move through a field with no supply and you have an eddy-current brake — the kind fitted to trains, which needs no contact and no power, because the motion generates the very current that stops it. A GMAW pool in an external field is a motor. A pool with a static field and no welding current is an eddy-current brake.

The proof, in one line of energy bookkeeping

The Lenz argument above is the intuition. Here is the same thing as an identity. Power delivered to the fluid per unit volume is P = F · u. For the two cases:

```
stirring P = (J_arc x B) . u sign depends on geometry
braking P = (- sigma B^2 u) . u = - sigma B^2 |u|^2
```

The braking expression is a negative constant times the square of a magnitude. It is **less than or equal to zero for every u, every B, and every conductivity**. There is no configuration, no field orientation and no flow pattern that makes it positive. The stirring expression has no such constraint — its sign is set by geometry, and arranging for it to be positive is exactly what an electromagnetic stirrer is for.

And the energy is fully accounted for. The induced current has magnitude sigma·u·B, so it dissipates Joule heat at

```
|J|^2 / sigma = (sigma u B)^2 / sigma = sigma B^2 u^2
```

THE BOOKKEEPING CLOSES EXACTLY

At 20 mT and 0.3 m/s, the braking force removes kinetic energy at **131.4 W/m^3** and the induced current dissipates Joule heat at **131.4 W/m^3**. Not approximately — the two expressions are algebraically identical.

Every joule the field takes out of the flow reappears as heat in the melt. Nothing is stored and nothing goes anywhere else. That is what makes this a *strictly dissipative* operator rather than a force in the ordinary sense.

What is actually happening to the electrons

Both cases come down to the same mechanism — electrons are deflected sideways and hand that transverse momentum to the ion lattice, which *is* the bulk liquid. What differs is what set the electrons moving in the first place, and therefore who pays.

	INJECTED	MOTIONAL
What starts the electrons moving	An external EMF from the power supply drives them through the metal at a drift velocity	The whole melt — electrons and ions together — is carried along at u

	INJECTED	MOTIONAL
The magnetic deflection	Each electron feels a sideways force in B and scatters off the lattice, pushing the liquid	Electrons are pushed *relative to the lattice*, which is what constitutes the induced current
Who pays for it	The power supply, which must work harder against the back-EMF	The flow's own kinetic energy — there is nobody else to ask
Net effect on the melt	**Gains** momentum in a direction you choose	**Loses** the momentum it already had

Direction: why the field orientation column exists

In WAAM the current runs from the wire tip down through the pool into the substrate, so J is predominantly along -z. Crossing that with each field orientation gives a different force, and this is the entire reason the case matrix has a field-direction column:

FIELD ORIENTATION	RESULTING FORCE	CONSEQUENCE FOR THE DOMAIN
B along y (transverse, across the weld)	J × B along **x** — along or against the travel direction	Drives longitudinal flow. **Preserves** the XZ symmetry, so a half domain remains valid
B along x (along the travel direction)	J × B along **y** — sideways, across the bead	**Destroys** the symmetry. A half domain gives a physically meaningless result that still renders plausibly
B along z (vertical, parallel to the main current)	Zero from the axial current — but the arc current also spreads radially, and J_r × B_z is **azimuthal**	Drives swirl about the arc axis. Also **destroys** the symmetry

Compare this with the braking table earlier in this section. There, field orientation chose which existing circulation was damped; here it chooses the direction of a force that was not there before. That is the difference between selecting and directing.

Two consequences that shape the case matrix

	STIRRING (injected)	BRAKING (motional)
Scaling with B	J is fixed by the power supply, so the force goes as $\mathbf{B^1}$	J = sigma u B already contains one factor of B, so the force goes as $\mathbf{B^2}$ — and as u, which is what makes it a drag
Can it be steered?	**Yes.** The force points wherever J × B points, so rotating the field rotates the push. **This is why the case matrix varies field direction at all** — it only means something if there is injected current	**No.** The force is always along -u. Rotating B changes only *which components* get damped

That second row has a detail worth keeping. When u is parallel to B the two terms in the expansion cancel exactly and the force is zero: flow along the field is untouched while flow across it is damped. That asymmetry is what drives strong-field MHD flows towards quasi-two-dimensional structure. Different powers of B are also the experimental fingerprint of which mechanism is operating — a measured effect scaling linearly in B is current-driven, one scaling quadratically is induction.

Why this case is 490 times one-sided

Since both forces cross the same B, the force ratio is simply the current ratio:

CURRENT	MAGNITUDE
Driven through the pool by the arc	$\mathbf{1.07 \times 10^7\ A/m^2}$
Generated by the pool's own motion, sigma · u · B at 20 mT	$\mathbf{2.19 \times 10^4\ A/m^2}$

CURRENT	MAGNITUDE
Ratio	**489 ×**

THE WHOLE ARGUMENT IN ONE LINE

The induced current is feeble because the melt outruns the field. **The pool turns over roughly 180 times before the field gets any grip on it** — Section 4 works out exactly what that means and where the boundary lies.

So the effect that experimenters observe at 20–50 mT **cannot be braking**. It has to be the current-driven term. And the current-driven term is precisely the one laserbeamFoam does not have.

4. Bulk magnetic braking, and where its threshold lies

Section 3 said braking is negligible here, and Section 10 quotes a threshold of 269 mT. This section says what the phenomenon is and where its boundary lies, because "the field is too weak" is not a statement a thesis can make without a number behind it.

What the phenomenon actually is

A melt pool is not still liquid. It circulates: Marangoni stress drags the surface outward from the hot centre, the liquid sinks at the cooler edges, returns along the bottom and rises again in the middle. In Zhao's simulation that closed loop runs at roughly 0.3 m/s and turns over about every 10 milliseconds.

Bulk magnetic braking is that entire loop slowing down. The mechanism runs in four steps:

- Every parcel of liquid moving across the field has its electrons deflected sideways, which drives a current.
- Those currents have nowhere to go but back through the melt — the solidified shell and the shielding gas conduct far worse — so they **close on themselves as eddy-current loops inside the liquid**.
- Every element of that eddy current sits in the same field, so J × B pushes back on the very liquid whose motion created it.
- The circulation slows, and the kinetic energy it loses reappears as Joule heat, exactly as Section 3 showed.

YOU HAVE SEEN THIS DEMONSTRATION

It is the same physics as a magnet dropped down a copper pipe, or the eddy-current brake on a rollercoaster: relative motion between a conductor and a field induces currents, and those currents oppose the motion that made them. No contact and no power supply required.

The difference here is that **the conductor is the melt itself**, so it brakes against its own circulation, throughout its own volume, rather than against an external magnet.

Why "bulk", and what it is being distinguished from

"Bulk" — or equivalently "core" — is standard MHD vocabulary for the interior of a flow, as opposed to the thin boundary layers at its edges. The distinction is not cosmetic: **different forces set the balance in the two regions**, so the same field can win in one and lose in the other. Putting the three competing terms of the momentum equation side by side at 20 mT settles it:

TERM	MAGNITUDE (N/m^3)	WHERE IT SETS THE BALANCE
Inertia rho U^2/L	79,500	Everywhere except thin layers at the edges — **this is the bulk**
Magnetic sigma B^2 U	**438**	—
Viscous mu U/L^2	41.7	Only in thin sheared layers against a boundary

BOTH STATEMENTS ARE TRUE AT ONCE, AND THAT IS THE WHOLE POINT

At 20 mT the magnetic force is **11 times stronger than viscosity** and **182 times weaker than inertia**. So the field *wins* wherever viscosity sets the balance, and *loses* wherever inertia does.

Viscosity sets the balance only in the boundary layers. Inertia sets it everywhere else. That is precisely why Ha = 3.2 and N = 0.0055 are not contradictory — they are the same field measured against two different opponents, in two different places. Ha is the ratio to viscosity; N is the ratio to inertia.

What you would actually see if the bulk were braked

- The circulation velocity falls, and the recirculation cells distort — squeezed, stretched along the field, or gone entirely.
- Convective heat transport drops, so the pool moves towards being conduction-dominated. Temperature gradients steepen and the pool changes shape.
- **And therefore the microstructure changes**: weaker stirring means less dendrite fragmentation, which means coarser grains. This is the whole reason the thesis cares.

None of that happens at 5–100 mT with a static field and no injected current. What happens instead is a Hartmann layer — a thin region near the boundary where the field does beat viscosity and does modify the shear.

A HYPOTHESIS WORTH RECORDING NOW

One of those boundaries is **the solidification front**. That is where dendrites grow, where solute is rejected, and where the microstructure is actually decided.

So a field too weak to touch the bulk may still be modifying the shear in the layer where grain structure is set. That would be a **third mechanism**, distinct both from bulk braking and from current-driven stirring, and it is a candidate explanation for effects reported experimentally at field strengths that cannot possibly be braking the pool.

This is speculation, not a result. But it is testable with the machinery this thesis already has, and if it holds it is a more interesting finding than the one currently planned.

The interaction parameter is a force ratio

A force is never large or small on its own — only relative to what it competes with. In the core of a melt pool the competition is inertia: the flow's own tendency to keep going. Comparing the two:

```
braking force density ~ sigma u B^2
inertial force density ~ rho U^2 / L
```

```
N = sigma u B^2 / (rho U^2 / L) = sigma B^2 L / (rho U)
```

That ratio is the **magnetic interaction parameter** N, also called the Stuart number. It is not a property of the field alone — it is the field measured against the flow it is trying to stop.

Why "bulk", and not just "braking"

Because there are two competitions, in two different places, and they have different answers. Near a solid boundary the flow is slow and viscosity dominates; in the core the flow is fast and inertia dominates. So a magnetic field faces a different opponent in each region:

NUMBER	COMPARES THE MAGNETIC FORCE AGAINST	GOVERNS
Ha = B L sqrt(sigma/mu)	Viscous force	The **boundary layers**. Ha > 1 means Hartmann layers form, of thickness roughly L/Ha
N = sigma B^2 L / (rho U)	Inertial force	The **bulk**, or core. N > 1 means the field restructures the main circulation itself

The two are not independent — N = Ha^2 / Re, so any two of the three fix the third. But they answer different questions, and conflating them is how a field that visibly alters a boundary layer gets reported as though it had stirred the pool.

> **THIS IS THE DISTINCTION THAT MATTERS FOR YOUR RESULTS**
>
> At 20 mT this problem has **Ha = 3.2** and **N = 0.0055**. Both are true at once. The field is strong enough to form Hartmann layers at the solid boundary, and roughly two hundred times too weak to touch the circulation in the middle.
>
> So "does the magnetic field affect the melt pool?" has no single answer at this field strength. It affects the boundary layer and not the bulk. Any reported effect must say which one it is claiming.

The four regimes

REGIME	CONDITION	WHAT THE FLOW DOES
Nothing	N << 1, Ha < 1	The field is too weak to matter anywhere. Streamlines are as they would be with no field at all
Boundary layers only	N << 1, Ha > 1	Hartmann layers form at the walls. The core circulation is unchanged. **Magnetised WAAM at 5-100 mT sits here**
Bulk restructured	N ~ 1	The magnetic force is comparable to inertia. Circulation slows, recirculation cells change shape or disappear. **This is the threshold**
Magnetically dominated	N >> 1	Inertia is negligible. Flow aligns with the field and becomes quasi-two-dimensional; Joule dissipation dominates the energy budget

Where the 269 mT comes from, and how much it can move

Setting N = 1 and solving for the field:

```
B_threshold = sqrt( rho U / (sigma L) )
```

The threshold **rises** with density and velocity — a heavier, faster melt is harder to stop — and **falls** with conductivity and pool size, because a more conductive or larger melt gives the induced current more to work

with. Since U and L are not known precisely, the honest thing is to show how far the answer moves:

U (m/s)	L (mm)	B FOR N = 1 (mT)	CORRESPONDS TO
0.1	3.0	**156**	slow circulation
0.3	3.0	**269**	the working estimate
0.9	3.0	**467**	droplet-impact velocity
0.3	1.0	**467**	a small first-layer pool
0.3	5.0	**209**	a wide pool, later layers
0.3	8.0	**165**	the widest measured, 7.76 mm

THE CONCLUSION IS ROBUST TO EVERY INPUT

Across the whole plausible range of pool velocity and size, the bulk braking threshold lands between **150 and 470 mT**. The published magnetised-welding literature works at **5–100 mT**.

So the finding does not depend on picking the right U or the right L. Whatever reasonable values you choose, static-field bulk braking is out of reach by an order of magnitude — and any observed effect in that literature has to have another cause.

The same threshold, read as a race between two clocks

N can be rewritten as a ratio of times, which is the most physical way to hold it:

```
N = sigma B^2 L / (rho U) = (L/U) / (rho / sigma B^2) = t_flow / t_magnetic
```

CLOCK	AT 20 mT	AT 269 mT	WHAT IT TIMES
t_{flow} = L/U	0.010 s	0.010 s	One circuit of the pool
$t_{magnetic}$ = rho / (sigma B^2)	1.8 s	**0.010 s**	The field appreciably slowing the melt

N = 1 is the field strength at which those two clocks tick together. At 20 mT the melt completes about 180 circuits before the field has slowed it appreciably — the flow simply outruns the braking. At 269 mT the field damps the flow in exactly the time the flow takes to go round once, which is what it means for the two forces to be comparable.

A MESH REQUIREMENT THAT FALLS OUT OF THIS

The Hartmann layer is roughly L/Ha thick. At 20 mT that is 925 micron and at 50 mT it is 370 micron — both comfortably resolved by the 250 micron cells in the case matrix. But by **100 mT the layer is 185 micron, thinner than one cell.**

So the high-field end of the magnetic sweep is under-resolved *in exactly the region where the only resolvable magnetic effect lives*. Either cap the sweep, or refine locally above roughly 60 mT. Worth deciding before the cases are queued rather than after.

5. This retires the Phase 1 puzzle

Phase 1 ran `Plate2D` with and without a 20 mT field and found no resolved effect on melt velocity. That was written up as "unresolved" — below the measurement's noise floor. The scaling says something better.

CONFIGURATION	HARTMANN NUMBER Ha	INTERACTION PARAMETER N
Plate2D as run, sigma = 10^6 S/m	0.28	7.6×10^{-4}
Plate2D at the true sigma = 3.65×10^6	0.54	2.8×10^{-3}

AN EXPECTED NULL, CORRECTLY REPRODUCED

With no current injection, Plate2D at 20 mT sits **four orders of magnitude** below the threshold for bulk magnetic braking. The Phase 1 result was not a resolution failure. It is the physically correct answer for that configuration.

Re-running at the true conductivity would not change it: conductivity buys a factor of 3.65 against a shortfall of 10^4. **Gate B proved the machinery works. It could never have shown a magnetic effect on the melt, and no field strength would have made it.**

Revise the Phase 1 write-up accordingly. "Expected null, correctly reproduced" is a stronger claim than "unresolved", and it is the true one.

6. The new number: electrical conductivity

The paper never needed this. Zhao computes the Lorentz force from an *assumed* Gaussian current distribution (their Eqs 22–24, after Kumar & DebRoy 2003) and never invokes Ohm's law. So electrical conductivity appears nowhere in Table 4, and it cannot be cited to them. It is new work — and it sets both the Lorentz force and the Joule heating.

Nobody has measured it for Al–5%Mg. It had to be constructed:

```
SOLID   300-815 K   rho = 1e-8(0.1743 + 7.502e-3 T + 4.028e-6 T^2) + 3.30e-8  Ohm.m
MUSHY   815-906 K   linear in liquid fraction
LIQUID  906-2000 K  rho = 1e-8(8.2019 + 1.9799e-2 T - 2.1227e-6 T^2) + 3.0e-8
```

Giving sigma = 1.64×10^7 S/m at 300 K, 8.15×10^6 at the solidus, **3.65×10^6 S/m at the liquidus**, and 2.36×10^6 at 2000 K.

How it was built, and how far each piece can be trusted

PIECE	BASIS	CONFIDENCE
Solid backbone	Desai, James & Ho (1984) pure-aluminium recommended values, plus a temperature-independent Matthiessen offset set by the measured 29% IACS of 5083/5356	**Good.** Cross-checked against Brandt & Neuer's measured AlSiCu-19, an alloy of matching room-temperature resistivity: **agrees to 1%**
Melting jump, factor 2.19–2.42	Brandt & Neuer (2007) measured this for eleven aluminium alloys. Band width only 5%, and no trend with composition	**Fair.** Transferred from Al-Si and Al-Si-Cu to Al-Mg. **Nobody has measured Al-Mg.** State this as a limitation
Liquid branch	Two independent routes: jump-ratio from the solidus, and pure-liquid aluminium plus a solute offset	**Weakest.** The two agree to 8%, but both are constructions. 15% at the liquidus, widening to 25% by 2000 K

A CORRECTION TO CARRY FORWARD

The MTHD tutorial uses sigma = 10^6 S/m for the metal. That **understates the real melt by 3.65 times** — and therefore understates the Hartmann number by 1.91 and the interaction parameter by 3.65. Any tutorial-derived intuition about field strength is biased low and should not be carried into the case matrix.

7. Thermal conductivity came free — and validates the above

The Wiedemann–Franz law ties thermal conductivity to electrical conductivity through $k = L_0 \cdot \sigma \cdot T$, with the Sommerfeld value $L_0 = 2.44 \times 10^{-8}$. Since Mills' tables are paywalled, this was the route to k(T). It is also a test: three independent checks where the answer is already known.

CHECK	WIEDEMANN-FRANZ	MEASURED	ERROR
Pure aluminium, solid at the melting point	213	211	**+1.0%**
Pure aluminium, **liquid** at the melting point	91.7	91	**+0.8%**
Al–5%Mg at 300 K	120.2	123	**–2.2%**
Pure aluminium at 300 K — the one that misses	263	237	+11%

All values in W/m/K.

The one failure is the one that should fail. In a *pure* metal at intermediate temperature, inelastic phonon scattering breaks Wiedemann–Franz and the effective Lorenz number dips below L_0. In a concentrated alloy, elastic impurity scattering dominates and the law holds. The pattern — pure metal off by 11% at 300 K, alloy right to 2%, both materials right to 1% at high temperature — is the physics behaving, not a coincidence. The agreement runs both ways: it is also evidence that the conductivity construction in Section 6 is sound.

ONE HONEST CAVEAT ON THE LIQUID BRANCH

International Journal of Thermophysics 40 (2019) reports that the ideal Lorenz number **overestimates** thermal conductivity for liquid Al-Si alloys. The liquid k values here should be treated as an upper bound, not a best estimate.

8. The property set

Thermal conductivity rises through the solid — the opposite of pure aluminium, because a concentrated alloy sits in the impurity-scattering regime — then **halves on melting**, from 162 to 81 W/m/K. Specific heat is anchored on pure aluminium; the Neumann–Kopp correction for 5% magnesium shifts it by 0.7%, well inside the uncertainty. Viscosity follows Assael et al. (2006) for liquid aluminium, which gives the accepted 1.25 mPa·s at the melting point.

T (K)	PHASE	σ_e (S/m)	k (W/m/K)	c_p (J/kg/K)	mu (Pa.s)
300	solid	1.643e+07	120.2	900	—
400	solid	1.405e+07	137.1	946	—
500	solid	1.215e+07	148.2	992	—
600	solid	1.061e+07	155.3	1037	—
700	solid	9.346e+06	159.6	1083	—
800	solid	8.296e+06	161.9	1129	—
815	solid	8.154e+06	162.2	1135	—

T (K)	PHASE	sigma$_e$ (S/m)	k (W/m/K)	c_p (J/kg/K)	mu (Pa.s)
860	mushy	5.927e+06	124.4	1157	—
906	liquid	3.650e+06	80.7	1180	1.334e-03
1000	liquid	3.463e+06	84.5	1180	1.085e-03
1200	liquid	3.134e+06	91.8	1180	7.798e-04
1500	liquid	2.768e+06	101.3	1180	5.602e-04
1800	liquid	2.502e+06	109.9	1180	4.493e-04
2000	liquid	2.364e+06	115.3	1180	4.024e-04

Full tables at 25 K resolution are in `alMg5_properties.csv`.

9. Three problems found in Zhao's Table 4

Reading a table closely enough to retype it is a different act from reading it to understand the paper. Three things surfaced that affect the replication.

1. THE SURFACE TENSION IS PURE ALUMINIUM

The pair 0.85 N/m and dgamma/dT = –1.55 × 10^{-4} N/m/K is footnoted to Mills reference **[34], the pure-aluminium chapter** — while every thermal property in the same table comes from reference [33], the **LM5/5182 alloy** chapter. It is the paper's one deliberate pure-aluminium substitution.

Magnesium is strongly surface-active in aluminium, so a true Al–5Mg value is lower. The 0.85 figure happens to coincide with the *oxidized* pure-aluminium value (0.865, Garcia-Cordovilla 1986); clean pure aluminium is 1.10 N/m. No measured Al–5Mg value could be located anywhere.

Marangoni convection drives the paper's headline surface-flow result. This is therefore the largest single uncertainty in the replication, and it must be carried as a sensitivity case rather than inherited silently.

2. THE ARGON DENSITY IS OFF BY FOUR ORDERS OF MAGNITUDE

Zhao's Section 3.3 states the plasma density as 6 × 10^{-6} kg/m^3 for the drag-force term. Argon is 1.62 kg/m^3 at 300 K and roughly 0.05 even at 10,000 K. Almost certainly a typographical error; the consequence is that plasma drag on the droplet is understated. Flagged in the dictionary, **not silently corrected** — the replication should reproduce what the paper did, with the discrepancy recorded.

3. THE MUSHY-ZONE CONSTANT IS NEVER STATED

Zhao gives only the conventional range, 10^4 to 10^7, and never says which value was used. A_{Darcy} is therefore a **free parameter of your replication**, not a value inherited from the paper. Set to 10^6 here and flagged for the sensitivity cases. Do not present it as the paper's number.

And one modelling limitation worth recording

Table 4 uses a single constant density, 2650 kg/m^3, for both phases, with the Boussinesq approximation carrying only thermal expansion. Real liquid Al–5Mg near the liquidus is about 2350 kg/m^3, so the melt is

modelled roughly 13% too heavy. That biases every inertial term. Kept at 2650 for replication fidelity, recorded in the dictionary so it is not forgotten.

10. Scaling results that set the case matrix

QUANTITY	RESULT	WHAT IT MEANS
Ha = 1	at **6.2 mT**	Hartmann boundary layers begin to form
N = 1	at **269 mT**	Order-unity bulk braking. **No experiment in the literature reaches it.**
Magnetic Reynolds number	4.1×10^{-3}	Much less than 1: the induced field is negligible and the quasi-static approximation is safe
J × B against gravity	equal at **2.4 mT**	The applied field beats gravity almost immediately
J × B against the arc's own self-field	equal at **13.5 mT**	The defensible crossover figure. Supersedes the earlier estimate of 20–30 mT
Joule heating	3.6 W against 1512 W	**0.24%** of the arc. Negligible — but now quantified with a real conductivity instead of asserted

The force ledger, compared honestly

Surface forces enter the momentum equation as stresses. The continuum-surface-force conversion to a volumetric force divides by the interface thickness, so any comparison in N/m^3 is mesh-dependent and therefore meaningless. Body forces are put on the same footing by multiplying by the pool length scale:

FORCE	KIND	STRESS (Pa)	RELATIVE TO GRAVITY
Arc self-field J × B	body	435.09	5.58 ×
Surface tension	surface	283.33	3.63 ×
Arc pressure, peak	surface	227.81	2.92 ×
Gravity	body	77.99	1.00 ×
Marangoni	surface	15.50	0.20 ×
Buoyancy (Boussinesq)	body	0.47	0.01 ×

For scale: an external field of 20 mT contributes 645 Pa — 8.3 times gravity, and larger than every other term in the table.

11. What is still open

- **Mills (2002), pages 32–36, is paywalled.** Specific heat and thermal conductivity here are constructions, not Mills' tables. One interlibrary-loan request would settle them — and would also reveal whether Mills lists electrical resistivity for 5182, which would collapse most of the uncertainty in Section 6.
- **Joshi et al. (2013), *Advanced Materials Research* 665:76** — "Electrical Resistivity of Liquid Al-Mg Binary Alloys". Theory only, and paywalled, but it is the closest primary source to this exact system that exists.
- ***Metals* 10(11):1447, Table 1** — EN AW 5754 under an external field, and it names its property sources. The most directly comparable published work found.
- **Is Zhao 2022's field static or oscillating?** If it oscillates, a third mechanism — induction — applies, and the scaling in Sections 2 to 4 needs a shielding-parameter analysis before the validation target can be used.

- **No measurement of liquid Al-Mg resistivity exists at any composition.** This belongs in the thesis as a stated limitation, not buried in a dictionary comment.

Immediate next step

The dictionary uses its own key names. Before Phase 3 they must be mapped onto what laserbeamFoam actually reads, by listing the tutorial dictionaries rather than guessing. Then Phase 3 proper: Goldak's double ellipsoid, arc pressure and surface heat loss as coded sources, and the non-chaotic parallel regression case.

Sources

SOURCE	WHAT IT PROVIDES	LOCATOR
Desai, James & Ho (1984)	*J. Phys. Chem. Ref. Data* 13(4):1131. Electrical resistivity of aluminium. CINDAS recommended values	`srd.nist.gov/jpcrdreprint/1.555725.pdf`
Brandt & Neuer (2007)	*Int. J. Thermophysics* 28(5):1429. Resistivity and conductivity of aluminium alloys through melting	`doi 10.1007/s10765-006-0144-0`
Garcia-Cordovilla, Louis & Pamies (1986)	*J. Materials Science* 21:2787. Surface tension of pure aluminium and Al-Mg	`doi 10.1007/BF00551490`
Joshi et al. (2013)	*Adv. Materials Research* 665:76. Electrical resistivity of liquid Al-Mg binaries — theory	`doi 10.4028/www.scientific.net/AMR.665.76`
Int. J. Thermophysics 40 (2019)	Thermal and electrical conductivity of liquid Al-Si alloys. Source of the Lorenz-number caveat	`doi 10.1007/s10765-019-2497-1`
Dinsdale & Quested (2004)	*J. Materials Science* 39. Viscosity of aluminium and its alloys	`doi 10.1023/B:JMSC.0000048735.50256.96`
Mills (2002)	*Recommended Values of Thermophysical Properties for Selected Commercial Alloys*. Zhao's refs [33] Al-LM5/5182 and [34] pure aluminium	`doi 10.1533/9781845690144`
Zhao, Wei, Long, Chen, Liu & Ou (2021)	*Welding in the World* 65:1571–1590. The replication target	`doi 10.1007/s40194-021-01123-1`
NDT conductivity chart	Aluminium alloy conductivity in %IACS. Source of the 29% figure for 5083/5356	`content.ndtsupply.com`